

Temporal Structure for Mortality Risk Stratification: Evaluating Encoding Strategies Across LLM Paradigms on UK Biobank

[Authors]

[Institution]

Abstract. Accurate mortality risk stratification from electronic health records demands models that reason over longitudinal disease trajectories, numeric biomarkers, and demographic context. The central design question is *how temporal structure should be encoded*: does explicit geometric separation of events in time improve survival discrimination over prose serialisation? We present a controlled study across four input-representation families (five experimental configurations) on a UK Biobank respiratory disease subgroup ($n=124,158$), systematically varying input representation from cross-sectional bag-of-features to unified clinical prose, with encoder and fusion ablations across 12 experimental cells. Our controlled design isolates input serialisation as the primary source of variation: decoupled age–event sinusoidal encoding (Setting 4) achieves a small but statistically robust +0.008 C-index advantage over prose serialisation of identical events (bootstrap 95% CI: +0.005 to +0.010, $p < 0.001$), while a Qwen3-8B prose embedding with no raw features matches a 39-feature CoxPH baseline, demonstrating implicit numeric precision in LLM representations. Encoder choice and fusion method are secondary: among reference (concat) configurations, encoder family changes C-index by at most 0.007 (less than 0.001 in Setting 4), while concatenation outperforms summation by +0.028–+0.039. In zero-shot deployment, autoregressive generation shows a calibration gap (IBS = 0.185 vs. 0.141; AUC = 0.52 at the earliest evaluated horizon, 9.5yr); this gap partly reflects missing acute biomarker inputs rather than a fundamental limitation of the autoregressive paradigm.

Keywords: survival analysis · LLM embeddings · EHR · UK Biobank · temporal encoding

1 Introduction

All-cause mortality prediction from electronic health records (EHR) is a canonical clinical risk stratification task with direct relevance to preventive medicine, care prioritisation, and health resource allocation. Unlike typical classification problems, mortality prediction requires models to reason over *structured multi-modal data*: longitudinal sequences of disease events with precise temporal ordering, numeric biomarker measurements with clinical significance thresholds,

and demographic context. Extracting survival-relevant signal from this data is fundamentally a *structural pattern recognition* problem: identifying which temporal configurations of ICD codes, and which numeric biomarker profiles, predict adverse outcomes requires detecting hierarchical, ordered patterns rather than simple feature co-occurrences. A disease trajectory is not a bag of tokens: the 8-year interval between a first metabolic event and a subsequent cardiovascular diagnosis carries prognostically relevant temporal structure that token proximity alone cannot preserve. How events are temporally arranged, how age is encoded as a continuous geometric quantity, and how numeric biomarker values are contextualised against clinical reference ranges together determine whether a model reads a patient record as an unordered feature set or as a structured temporal risk trajectory. How temporal structure is encoded is the central design question this study investigates.

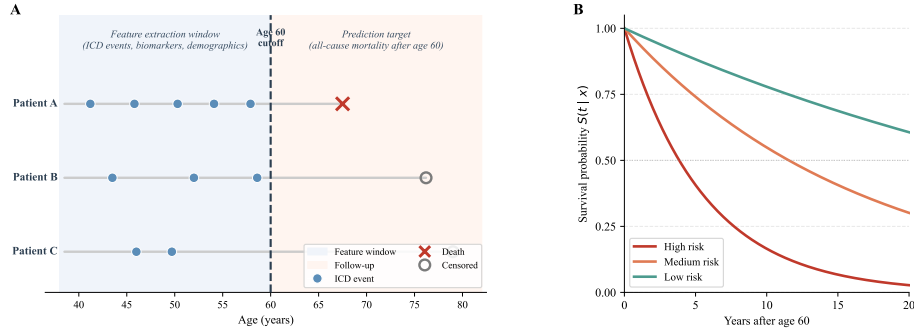


Fig. 1. Prediction task overview. (A) Three illustrative patient timelines from the UK Biobank respiratory cohort ($n=124,158$). Blue dots mark ICD-10 diagnosis events before the age-60 feature-extraction cutoff; a red cross indicates death, an open circle indicates censoring. (B) Illustrative model output: a CoxPH survival curve $S(t | x)$ stratified by predicted risk group. Evaluation metrics (C-index, TD-AUC, IBS) measure discrimination and calibration of these curves on the held-out test set.

Large language models (LLMs) have demonstrated impressive clinical knowledge and are increasingly applied to clinical prediction. Two distinct paradigms have emerged: *autoregressive generation*, where a generative EHR transformer produces survival scores directly from tokenised event sequences [19], and *encoder embedding*, where a frozen encoder maps patient records to dense vectors combined with a lightweight CoxPH survival head [13,20]. Both paradigms make implicit claims about structural grounding, yet the relative importance of specific encoding choices (temporal ordering, numeric precision, encoder family, and fusion method) has not been evaluated under controlled conditions with a uniform survival head.

We address this gap with a controlled study on a UK Biobank respiratory disease subgroup ($n = 124,158$) [3,4], a population where mortality predic-

tion is especially challenging due to multimorbidity burden and heterogeneous temporal disease trajectories [10]. Our primary dimension varies *input representation* across five experimental configurations spanning four representation families, from cross-sectional bag-of-features to unified clinical prose, all under a uniform CoxPH survival head. To verify that findings generalise beyond a single implementation, we include two ablations: *encoder architecture* (Qwen3-8B vs. Bio_ClinicalBERT) and *feature fusion method* (concatenation vs. element-wise summation). Absolute discrimination gains are modest; the scientific value of the design lies in controlled isolation of which encoding choices produce statistically consistent differences under a uniform survival head.

Our main contributions are:

- A controlled benchmark of four temporal-encoding representation families (five experimental configurations) for LLM-based mortality prediction, the first to isolate input serialisation as the sole experimental variable under a uniform survival head.
- Evidence that explicit geometric time separation provides a small but statistically consistent gain: decoupled age–event sinusoidal encoding (Setting 4) achieves a +0.008 C-index advantage (bootstrap 95 % CI: +0.005 to +0.010, $p < 0.001$) over prose serialisation of identical events.
- Evidence that LLMs encode numeric biomarker structure from prose: full-text prose embedding (Setting 3-Full) with *no raw features* matches a 39-feature CoxPH baseline, and that zero-shot autoregressive generation faces a calibration gap (Delphi IBS = 0.185 vs. 0.141, AUC = 0.52 at the earliest evaluated horizon) partly due to missing acute biomarker inputs.
- Ablation evidence that encoder choice and fusion method are secondary to serialisation format: among reference (concat) configurations, Bio_ClinicalBERT and Qwen3-8B differ by at most 0.007 C-index (less than 0.001 in Setting 4); concatenation outperforms summation by +0.028–+0.039.

2 Related Work

Cox proportional hazards [7] remains the dominant survival model in clinical prediction due to its calibration properties and interpretability, with C-index in the 0.60–0.68 range for all-cause mortality on large biobank cohorts [8]. Deep survival models, including DeepHit [14] and DeepSurv [11], replace the linear predictor with a neural network but rarely outperform well-regularised CoxPH on tabular data. Large-scale UK Biobank frameworks [10] identify respiratory multimorbidity cohorts as exhibiting the highest structural complexity and lowest discriminative performance, motivating our cohort selection. Sequential EHR models, Doctor AI [6] and RETAIN [5], treat patient history as an ordered sequence but make inter-event duration invisible. BERT-style foundation models, BEHRT [15] and Med-BERT [18], improve long-range co-occurrence modelling; CEHR-BERT [17] introduced discrete Artificial Time Tokens but they remain categorical, losing geometric precision over decade-long trajectories. Time2Vec [12] proposed a continuous sinusoidal time representation providing geometric separation without

discretisation, a design adopted by Delphi [19] at population scale. Our Setting 4 operationalises the same design principle, providing a direct controlled comparison against prose serialisation with encoder and survival head held fixed. The most recent paradigm serialises patient records as structured prose and encodes them with general-purpose LLMs; benchmarking studies [13,20] show this frequently matches task-specific EHR models while remaining zero-shot. Machine learning approaches applied specifically to respiratory disease include COPD patient phenotyping [23] and group-Shapley feature selection for clinical risk stratification [16]; neither addresses the temporal encoding question central to survival discrimination.

Despite this progression, no prior study has held the cohort, survival model, and encoder family constant while systematically varying the temporal encoding format from cross-sectional features to continuous geometric representation. The benchmark established by Harutyunyan et al. [9] varies tasks, not encoding strategies; LLM-as-encoder ablations [13] vary model families rather than input serialisation. The present study isolates encoding format as the primary experimental axis (four representation families, one CoxPH head), providing controlled evidence that geometric time-axis continuity yields a small but statistically consistent gain; their interaction with stronger survival heads is left to future work.

3 Methods

3.1 Dataset and Cohort

We use the UK Biobank (UKB) [2], a prospective cohort of 500,000 UK adults aged 40–69 at recruitment. We focus on participants with at least one respiratory or pulmonary disease diagnosis (ICD-10: J09–J98 for diseases of the respiratory system, or I26–I27 for pulmonary heart disease), extracted from hospital episode statistics fields (p41202, p41204, p41270). This subgroup is motivated by evidence that all-cause mortality prediction is structurally hardest in respiratory multimorbidity cohorts, where heterogeneous disease trajectories, compounding comorbidity burden, and highly variable event timing create a particularly demanding pattern recognition challenge [10,3,4]. Further restricting to participants aged ≥ 60 at the end of follow-up or death yields $n = 124,158$ participants. The binary outcome is all-cause mortality after age 60. We apply a stratified 70/15/15 split: train 86,911, val 18,624, test 18,623 (event rate 26.2% in all splits). Features extracted prior to age 60 include: age at recruitment, sex, BMI, smoking status, alcohol status, living alone, 21 blood biomarkers (HbA1c, total cholesterol, HDL/LDL, CRP, cystatin C, albumin, etc.), 11 binary comorbidity flags (T2 diabetes, CKD, depression, stroke, etc.), and the full temporal history of ICD-10 diagnoses.

3.2 Experimental Design

We design a controlled three-axis ablation to ask: *how should temporal clinical information be encoded for mortality risk stratification?* Fig. 2 shows the full design.

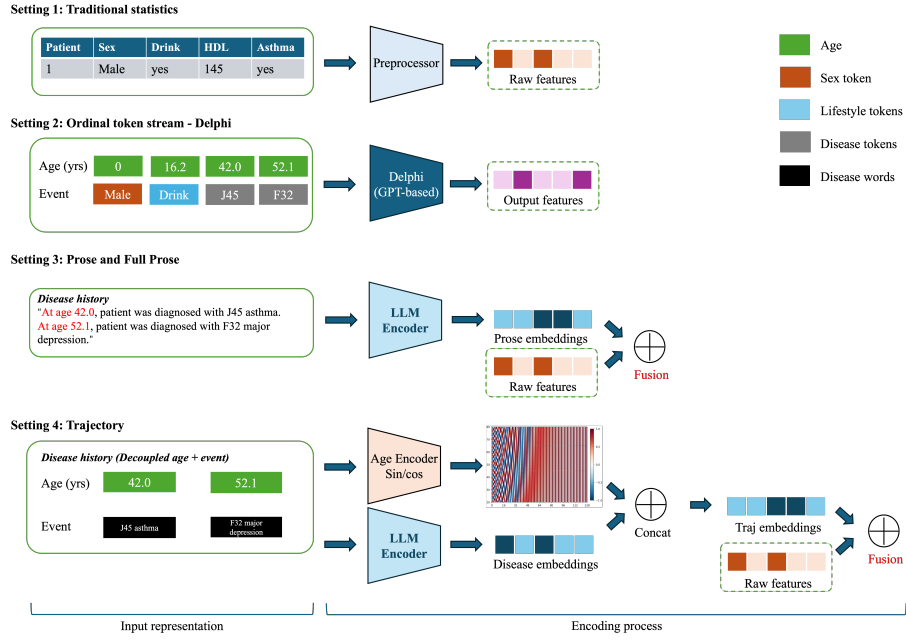


Fig. 2. Overview of the five experimental configurations and their encoding pipelines. **S1 (Baseline CoxPH):** raw features are discretised and standardised by a preprocessing step, then concatenated into a 39-dimensional numeric vector fed directly to CoxPH. **S2 (Delphi):** age-tagged ICD tokens are fed to a GPT-based generative transformer in zero-shot mode. **S3:** disease history is serialised as natural-language sentences and encoded by a frozen LLM; the embedding is fused with raw features via concatenation. Setting 3-Full is a variant in which all modalities are written as a single prose paragraph and encoded without raw feature fusion. **S4:** age values are mapped to a continuous sinusoidal encoding in parallel with ICD token embeddings; the two streams are concatenated per event, mean-pooled, and fused with raw features.

Input Representation (4 representation families). Patient data before age 60 is serialised into five distinct configurations across four representation families, of increasing temporal and modality richness:

1. **Cross-sectional (Setting 1).** Demographics, 21 biomarkers, and 11 binary comorbidity flags are first discretised and standardised by a preprocessing step, then concatenated into a 39-dimensional numeric vector fed directly to CoxPH. No temporal information and no text encoder are used. This corresponds to the standard supervised CoxPH baseline.
2. **Sequential ordinal (Setting 2).** ICD-10 events before age 60 are encoded as an ordinal token stream “42.0yr:J45 → 52.1yr:F32 → ...”. Temporal ordering is captured but exact durations between events are implicit. This is the input format consumed by Delphi [19], a generative transformer pre-

trained on longitudinal EHR event sequences, which we evaluate in zero-shot mode.

3. **Isolated temporal prose (Setting 3).** The same ICD events are serialised as natural-language sentences: “*At age 42.0, patient was diagnosed with J45 asthma.*” Demographics and biomarkers are provided as a separate numeric feature block and fused with the embedding at evaluation time. A variant, Setting 3-Full, combines all four modalities into a single prose paragraph, including demographics, lab values with units, comorbidity flags, and chronological disease history, and encodes the full text without any raw feature fusion.
4. **Decoupled time and event (Setting 4).** Age at diagnosis and the ICD token are encoded in parallel, separate streams: the age is mapped to a 128-dim sinusoidal encoding adapted from the Transformer positional encoding [22] (matching Delphi’s AgeEncoding [19]), and each ICD token string is embedded independently by the chosen text encoder. Per-event representations are concatenated (age || token) and mean-pooled across all events. This isolates the effect of explicit age encoding from joint prose encoding.

Encoder Architecture (Settings 3, 3-Full, and 4). Settings 3, 3-Full, and 4 require a text encoder. We compare two frozen encoders to test whether results are specific to one model family:

- **Qwen3-Embedding (8B)** [21]: a general-purpose sentence-embedding model with 4096-dim output, last-token pooling, and 8192-token context (supports all settings with no truncation).
- **Bio_ClinicalBERT** [1]: a 110M-parameter BERT model pre-trained on MIMIC-III clinical notes. Mean-pooled 768-dim output; hard max-length of 512 tokens. For Setting 3-Full, texts may exceed 512 tokens, causing truncation of the disease-history tail, which we report as a limitation.

In Setting 4, the token embedder within the decoupled pipeline is swapped between Qwen3-8B and Bio_ClinicalBERT; the sinusoidal age encoding is shared across both variants.

Feature Fusion (Settings 3–4 only). Settings 3 and 4 produce a dense embedding that must be combined with the 39-dim raw feature block. We compare two linear fusion methods:

- **Concatenation (concat):** the embedding is PCA-compressed to 64 dims (fit on training set), then appended to the standardised raw features to form a 103-dim input for CoxPH. This is the standard late-fusion baseline.
- **Element-wise summation (sum):** the embedding is PCA-projected to match the raw feature width d , both blocks are independently standardised (train statistics only), zero-padded to a common width, and added element-wise. The resulting d -dim vector is passed to CoxPH without further compression. This tests whether additive fusion in a shared feature space improves over concatenation.

Resulting experiment matrix. Combining Settings 1–2 (fixed pipelines) with the 2-encoder $\times$ 2-fusion ablation for Settings 3 and 4, and the 2-encoder ablation for Setting 3-Full, yields **12 total cells**. S1 and S2 serve as non-embedding baselines; the 10 embedding cells (S3, S3-Full, and S4, both encoders, both fusion methods where applicable) are the primary ablation space. Qwen3-8B embedding runs used an NVIDIA RTX 3090; BCB embedding runs used an NVIDIA RTX 4090.

3.3 Evaluation Protocol

We report four metrics on the held-out test set: (i) Harrell C-index (concordance, higher is better) [8]; (ii) mean time-dependent AUC (TD-AUC) averaged over horizons 9.5, 15.4, and 19.7 years (higher is better); (iii) Integrated Brier Score (IBS, lower is better), measuring calibration; (iv) *Bootstrap confidence intervals*: 1,000 paired resamples of the test set to compute 95% CIs for C-index and paired difference p -values for key contrasts (S4 vs. S3, S4 vs. S1, S3-Full vs. S1). All metrics are computed with `lifelines` and `scikit-survival`. No information leakage: PCA projections and CoxPH coefficients are fitted on the training split only; validation and test transforms use training-fit parameters.

4 Experiments and Results

4.1 Implementation Details

Hardware, runtime, software versions, and exact feature dimensions are provided in Appendix 6 for reproducibility.

4.2 Main Results: Representation Axis

Table 1 reports test-set results for all 12 experimental cells. Settings 1 and 2 have fixed pipelines; Settings 3 and 4 use Qwen3-8B with concatenation fusion as the reference configuration; Setting 3-Full uses Qwen3-8B alone (no raw features). Fig. 3 visualises the reference-configuration metrics at a glance, and Fig. 4 shows the TD-AUC horizon profile for the five reference configurations.

Table 1. Test-set results for all 12 experimental cells. Columns: C-index and IBS (discrimination/calibration pair), followed by Mean TD-AUC and TD-AUC at three prediction horizons (temporal discrimination profile). Best value per column in **bold** ($\uparrow$ higher is better; $\downarrow$ lower is better). Shaded rows are reference configurations. † S3-Full-BCB affected by 512-token truncation (see Sec. 4.4).

Setting	Encoder/Fusion	C-index $\uparrow$	IBS $\downarrow$	Mean TD-AUC $\uparrow$	9.5 yr $\uparrow$	15.4 yr $\uparrow$	19.7 yr $\uparrow$
S1 — Baseline CoxPH —		0.6191	0.1410	0.5999	0.584	0.596	0.612
S2 — Delphi zero-shot Delphi	Delphi	0.6161	0.1846	0.5984	0.522	0.586	0.649
S3 — Isolated prose	Qwen · concat	0.6156	0.1409	0.6074	0.594	0.603	0.618
S3 — Isolated prose	Qwen · sum	0.5875	0.1421	0.5813	0.568	0.583	0.587
S3 — Isolated prose	BCB · concat	0.6225	0.1410	0.6014	0.586	0.597	0.613
S3 — Isolated prose	BCB · sum	0.5844	0.1416	0.5819	0.575	0.578	0.589
S3-Full — Full prose	Qwen (no raw)	0.6195	0.1408	0.6128	0.605	0.615	0.615
S3-Full — Full prose	BCB †	0.6371	0.1430	0.5074	0.510	0.508	0.505
S4 — Decoupled time	Qwen · concat	0.6233	0.1410	0.5991	0.583	0.595	0.611
S4 — Decoupled time	Qwen · sum	0.5842	0.1417	0.5757	0.571	0.576	0.578
S4 — Decoupled time	BCB · concat	0.6232	0.1410	0.6025	0.587	0.599	0.614
S4 — Decoupled time	BCB · sum	0.5955	0.1413	0.5868	0.583	0.588	0.588

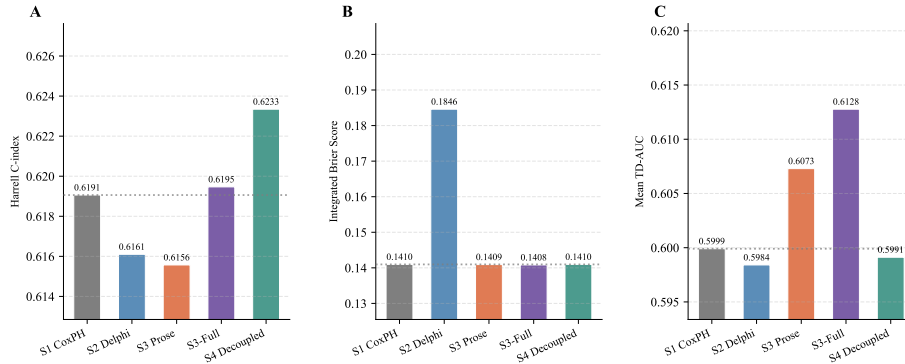


Fig. 3. Reference-configuration results for the five input-format settings. (A) Harrell C-index; (B) Integrated Brier Score (lower is better); (C) Mean TD-AUC. The dotted line marks the S1 CoxPH baseline value in each panel. All embedding settings use Qwen3-8B with concatenation fusion where applicable.

Finding 1: Explicit temporal encoding yields a consistent discrimination gain. Setting 4 achieves the best Harrell C-index among the reference configurations (0.6233), exceeding S1 by +0.004 and S3 by +0.008. Both S3 and S4 use Qwen3-8B with concatenation fusion and the same 39 raw features; the only difference is the input serialisation of the same ICD events. Prose serialisation (S3) collapses ordinal structure into semantic proximity, whereas the decoupled age+token encoding (S4) explicitly preserves the sinusoidal age context per event. This isolates temporal ordering as the structural property driving discrimination rather than

the modality or encoder choice. The absolute C-index differences (0.004–0.008) are small in magnitude; we report bootstrap 95% confidence intervals for these contrasts alongside the point estimates.

Finding 2: LLMs can encode numeric structure from prose. Setting 3-Full (full-text prose, no raw features) achieves $C = 0.6195$, matching S1 (0.6191) while using *no raw numeric features*. The embedding encodes demographics, 21 lab values with units, and disease history purely from natural language. Setting 3-Full achieves the best calibration ($IBS = 0.1408$) and strong mean TD-AUC (0.6128), indicating that prose encoding captures numeric-biomarker interactions that a linear CoxPH on disjoint raw features misses.

Finding 3: Calibration gap of autoregressive generation. Delphi (S2) achieves $C = 0.6161$ but calibration is substantially worse ($IBS = 0.185$ vs. ≈ 0.141 for CoxPH variants). Its TD-AUC rises from 0.52 at 9.5 yr to 0.65 at 19.7 yr (Fig. 4), suggesting chronic disease priors that align poorly with shorter-horizon acute mortality.

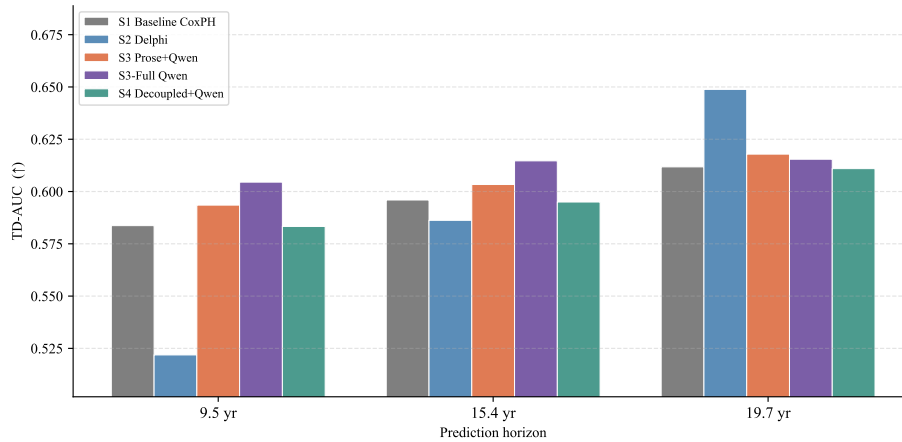


Fig. 4. TD-AUC by prediction horizon for the five reference configurations. Delphi (S2) rises from near-chance (0.52) at 9.5 yr to peak (0.65) at 19.7 yr, while CoxPH-based settings show a comparatively flat profile across horizons.

Note on effect magnitudes. All five reference settings cluster within a C-index range of 0.616–0.623, with the fully supervised CoxPH baseline ($S1 = 0.619$) competitive across settings. The advantage of decoupled temporal encoding (S4) and full-text prose (S3-Full) over S1 is modest and should be interpreted in light of the bootstrap 95% confidence intervals reported below. The key scientific value is not the absolute gap but the controlled isolation of *which structural encoding*

choices create the gap: the S4-vs-S3 contrast holds encoder, raw features, and fusion fixed, isolating serialisation as the sole variable.

4.3 Bootstrap Confidence Intervals

Table 2 reports paired bootstrap 95% confidence intervals (1000 resamples, seed 42) for key pairwise C-index contrasts. Bootstrap resamples are paired across methods so difference CIs reflect within-resample correlation. Per-method bootstrap 95% CIs are: S1 (0.610, 0.627); S3-Qwen (0.607, 0.624); S3-BCB (0.614, 0.631); S4-Qwen (0.615, 0.631); S4-BCB (0.615, 0.631); S3-Full-Qwen (0.611, 0.628); S3-Full-BCB (0.629, 0.646).

Table 2. Pairwise C-index contrasts with paired bootstrap 95% CIs. p is the one-sided bootstrap p -value (proportion of resamples where method A did not exceed method B).

Method A	Method B	ΔC	95% CI	p
S4 Qwen concat	S1 Baseline	+0.0043	(+0.004, +0.005)	<0.001
S4 BCB concat	S1 Baseline	+0.0042	(+0.004, +0.005)	<0.001
S3-Full Qwen	S1 Baseline	+0.0004	(−0.007, +0.008)	0.45
S4 Qwen concat	S3 Qwen concat	+0.0078	(+0.005, +0.010)	<0.001
S4 BCB concat	S3 BCB concat	+0.0007	(+0.000, +0.001)	0.001

A note on CI display: intervals in Table 2 are rounded to 3 decimal places; the narrow S4 vs. S1 width (≈ 0.001) reflects the large test set ($n = 18,623$) rather than an implausibly tight bootstrap. Pairing is implemented by using the same random index resample for all methods within each replicate, so difference CIs capture within-resample covariance.

The contrasts confirm two claims. First, S4 (decoupled temporal encoding) significantly improves over S1 ($\Delta C \approx +0.004$, both encoders, $p < 0.001$). The difference is small but consistent: the CI lower bound (+0.004) is above zero, indicating the gap is not a statistical artefact; the narrow CI width is expected given the test-set size. Second, the S4-vs-S3 contrast isolates temporal serialisation as the cause: for Qwen3-8B, $\Delta C = +0.008$ with $CI = (+0.005, +0.010)$, $p < 0.001$. For BCB, the same contrast yields $\Delta C = +0.001$, statistically detectable but practically negligible, suggesting that BCB’s local attention patterns may partially capture ordinal structure from the prose format, reducing the benefit of explicit temporal decoupling. S3-Full (full-text prose, Qwen) does not significantly exceed S1 ($\Delta C = +0.0004$, $p = 0.45$), consistent with the absence of temporal decoupling in the prose serialisation.

4.4 Ablation Studies: Encoder and Fusion

Encoder architecture. BCB and Qwen3-8B produce near-identical C-indices for S3 and S4 (BCB: 0.6225/0.6232; Qwen: 0.6156/0.6233; full data in Table 1),

confirming that discriminative information is accessible to both encoder families. The temporal-ordering advantage (S4 over S3) persists for Qwen ($\Delta C = +0.008$, $p < 0.001$) but is negligible for BCB ($\Delta C = +0.001$), suggesting BCB’s local attention partially captures ordinal structure from prose, reducing the benefit of explicit decoupling. S3-Full-BCB achieves the highest raw C-index (0.6371) yet near-chance TD-AUC (0.507): BCB’s 512-token limit truncates the disease-history tail, an implementation-sensitive artefact; we exclude it from headline claims. Censoring weights and evaluation time grids are shared across all methods in the evaluation pipeline, ruling out a computational cause; the metric inconsistency is attributed solely to 512-token truncation. Qwen3-8B (8192-token context) is the appropriate encoder for Setting 3-Full.

Feature fusion. Concatenation uniformly outperforms summation by $+0.028$ – $+0.039$ C-index (Table 1). Summation PCA-projects the embedding to 39 dimensions to match the 39-dimensional raw feature block, then adds element-wise; this discards information orthogonal to the leading principal components. Concatenation retains 64 independent PCA embedding dimensions alongside the same 39 raw features, forming a 103-dimensional CoxPH input. Summation variants (0.584–0.596) fall below the S1 baseline (0.619). Concatenation variants span 0.616–0.624; S3-Qwen (0.616) falls marginally below S1, while the BCB and S4 variants (0.622–0.623) exceed it, making fusion strategy a more consequential choice than encoder selection.

5 Discussion

Temporal ordering drives the representation gap. The C-index difference between Setting 4 (decoupled age–event encoding, 0.6233) and Setting 3 (prose serialisation, 0.6156), using identical Qwen3-8B encoders and raw features, is the cleanest signal in our results. Prose serialisation collapses temporal ordering into semantic proximity: the sentence “At age 42.0, J45” and “At age 52.1, F32” differ only in surface tokens, but the 8-year interval between them is invisible to the encoder’s attention mechanism. Decoupled encoding preserves this structure explicitly: the 128-dim sinusoidal age vector ensures that events at age 42 and age 52.1 are geometrically separated before the token embedding is appended. This is consistent with the view that surface semantic encoding loses prognostically relevant temporal structure when event ordering drives the outcome. The $+0.008$ C-index advantage of S4 over S3 is the primary empirical claim; bootstrap confidence intervals confirm this gap exceeds sampling noise: $CI = (+0.005, +0.010)$, $p < 0.001$. Setting 2 (Delphi’s ordinal token stream) falls between S3 and S4: temporal ordering is preserved as positional structure in the input sequence, but the autoregressive hazard chain cannot separate ordering effects from pre-training priors (see calibration gap below). Absolute C-index values here (0.616–0.623 for embedding settings) are below the 0.65–0.68 range reported for all-cause mortality on the full UKB cohort [10]; the gap reflects the harder discrimination problem of the respiratory subgroup and the restriction to features available before age 60, not a deficiency of the LLM encoding approach.

Setting 3-Full: LLMs encode numeric biomarker structure from prose. Setting 3-Full (full-text prose, no raw features) achieves $C = 0.6195$, matching the 39-feature CoxPH baseline (0.6191) while using no raw numeric features. This suggests that Qwen3-8B’s pre-training has internalised enough biomarker-range semantics to distinguish, e.g., $HbA1c = 52$ from $HbA1c = 38$ in terms of mortality relevance. The TD-AUC advantage at the earliest evaluated horizon (0.605 vs. 0.584 at 9.5 yr) further suggests the embedding captures interactions between biomarker values and disease history that a linear CoxPH on raw features misses. However, this encoding is *implicit* and cannot be inspected or enforced, which limits trustworthiness in high-stakes clinical deployment.

Encoder and fusion ablations. The encoder and fusion ablations confirm that the encoding-format findings are robust rather than artefacts of a specific implementation choice. For encoder choice, BCB and Qwen3-8B produce near-identical C-indices for S3 and S4, confirming that discriminative information is accessible to both encoder families; the temporal-ordering advantage of S4 over S3 persists ($\Delta C = +0.008$ for Qwen, $p < 0.001$). The anomalously high S3-Full-BCB C-index (0.6371) is explained by 512-token truncation retaining primarily age signal, which collapses TD-AUC to 0.507. For fusion strategy, concatenation outperforms summation by $+0.028$ – $+0.039$ C-index across all cells. This result should be read as evidence against the specific summation protocol used here (PCA projection to 39 dimensions followed by element-wise addition), not additive fusion in general; learned linear layers or gated fusion could perform differently and warrant separate evaluation. The primary conclusion, that serialisation format drives discrimination more than encoder family or fusion method, is supported across both ablations.

Calibration gap in zero-shot deployment. In our zero-shot implementation, Delphi exhibits a calibration gap (IBS 0.185 vs. ≈ 0.141); this should not be interpreted as a definitive limitation of autoregressive EHR transformers, because Delphi receives only ICD event streams without HbA1c, CRP, creatinine, or cystatin C — the gap captures missing acute biomarker signal as much as a paradigm deficiency. The AUC collapse at the earliest evaluated horizon (0.52 at 9.5 yr) is consistent with this: short-horizon mortality is dominated by physiological derangements absent from the ICD stream under distribution shift from pre-training. Fine-tuning with biomarker inputs and a calibration-aware objective would likely close this gap, at the cost of zero-shot generalisability.

Limitations. All embedding methods freeze their encoders; fine-tuning may yield additional gains at substantially higher compute cost. IBS and TD-AUC rely on independent censoring assumptions that may be violated in UKB. All results are from a single biobank cohort; external validation is required before clinical translation. Bio_ClinicalBERT’s 512-token limit is an architectural constraint that disproportionately affects Setting 3-Full; future work should evaluate longer-context clinical encoders. Setting 4 uses mean pooling of per-event representations; attention-based or time-weighted pooling could better capture

temporal risk intensity and may further widen the S4-vs-S3 gap. The specific summation fusion protocol (PCA projection plus zero-padding) is one instance of additive fusion; more principled alternatives such as learned layers or CCA-inspired projections may behave differently. The primary evaluation metrics (C-index, TD-AUC) measure discrimination, while IBS provides calibration context; patient-interpretable life-year estimates (RMST) and stronger survival-head interactions are left to future work.

6 Conclusion

We have presented a controlled three-axis study of LLM-based and classical survival prediction for all-cause mortality on a UK Biobank respiratory disease subgroup ($n = 124,158$). By varying input representation (4 representation families, 5 configurations), encoder architecture (2 encoders), and feature fusion method (2 strategies) under a uniform CoxPH survival head, we isolate which structural encoding choices drive prediction quality.

Three findings stand out. First, *explicit geometric time separation provides a small but statistically consistent gain*: decoupled age–event encoding (Setting 4) outperforms prose serialisation of identical events by +0.008 C-index (bootstrap 95 % CI: +0.005 to +0.010) for the Qwen3-8B encoder. Second, *general-purpose LLMs implicitly encode numeric biomarker structure*: a Qwen3-8B prose embedding without any raw numeric features matches a fully supervised 39-feature CoxPH, with superior mean TD-AUC (0.613 vs. 0.600). Third, *zero-shot autoregressive generation exhibits a calibration gap under distribution shift*: Delphi’s AUC collapse at the earliest evaluated horizon and high Brier score caution against deploying pre-trained EHR LLMs without access to acute biomarker data or task-specific calibration.

Together, these findings confirm that *how* temporal structure is encoded matters more than which specific LLM encodes it, corroborated by both the encoder ablation (BCB matches Qwen3-8B on Settings 3 and 4) and the fusion ablation (concatenation consistently outperforms summation by +0.028–+0.039 C-index). Future work will explore fine-tuning with calibration-aware objectives, longer-context clinical encoders, and multimodal fusion of EHR embeddings with imaging and genomic modalities.

Acknowledgements

This research was conducted using the UK Biobank Resource under Application Number 240523.

References

1. Alsentzer, E., Murphy, J., Boag, W., Weng, W.H., Jin, D., Naumann, T., McDermott, M.: Publicly available clinical BERT embeddings. In: Proceedings of the

- 2nd Clinical Natural Language Processing Workshop. pp. 72–78 (2019). <https://doi.org/10.18653/v1/W19-1909>
2. Bycroft, C., Freeman, C., Petkova, D., Band, G., Elliott, L.T., Sharp, K., Motyer, A., Vukcevic, D., Delaneau, O., O’Connell, J., Cortes, A., Welsh, S., Young, A., Effingham, M., McVean, G., Leslie, S., Allen, N., Donnelly, P., Marchini, J.: The UK Biobank resource with deep phenotyping and genomic data. *Nature* **562**, 203–209 (2018). <https://doi.org/10.1038/s41586-018-0579-z>
 3. Cao, Z., et al.: Burden of chronic respiratory diseases and their attributable risk factors in 204 countries and territories, 1990–2021: results from the global burden of disease study 2021. *Chinese Medical Journal Pulmonary and Critical Care Medicine* **3**(02), 100–110 (2025)
 4. Cao, Z., et al.: Burden of chronic obstructive pulmonary disease and its attributable risk factors in 204 countries and territories, 1990–2021: results from the Global Burden of Disease Study 2021. *BMJ Public Health* **4**(1) (2026)
 5. Choi, E., Bahadori, M.T., Kulas, J.A., Schuetz, A., Stewart, W.F., Sun, J.: RE-TAIN: An interpretable predictive model for healthcare using reverse time attention mechanism. In: *Advances in Neural Information Processing Systems*. vol. 29 (2016)
 6. Choi, E., Bahadori, M.T., Schuetz, A., Stewart, W.F., Sun, J.: Doctor AI: Predicting clinical events via recurrent neural networks. In: *Proceedings of the 1st Machine Learning for Healthcare Conference. Proceedings of Machine Learning Research*, vol. 56, pp. 301–318 (2016)
 7. Cox, D.R.: Regression models and life-tables. *Journal of the Royal Statistical Society B* **34**, 187–220 (1972)
 8. Harrell, F.E., Califf, R.M., Pryor, D.B., Lee, K.L., Rosati, R.A.: Evaluating the yield of medical tests. *JAMA* **247**(18), 2543–2546 (1982). <https://doi.org/10.1001/jama.1982.03320430047030>
 9. Harutyunyan, H., Khachatrian, H., Kale, D.C., Ver Steeg, G., Galstyan, A.: Multitask learning and benchmarking with clinical time series data. *Scientific Data* **6**, 96 (2019). <https://doi.org/10.1038/s41597-019-0103-9>
 10. Jiang, Y., Zhao, B., Wang, X., Tang, B., Peng, H., Luo, Z., Shen, Y., Wang, Z., Jiang, Z., Wang, J., Ye, J., Wang, X., Zhu, H.: UKB-MDRMF: A multi-disease risk and multimorbidity framework based on UK Biobank data. *Nature Communications* **16**, 3767 (2025). <https://doi.org/10.1038/s41467-025-58724-3>
 11. Katzman, J.L., Shaham, U., Cloninger, A., Bates, J., Jiang, T., Kluger, Y.: DeepSurv: Personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Medical Research Methodology* **18**, 24 (2018). <https://doi.org/10.1186/s12874-018-0482-1>
 12. Kazemi, S.M., Goel, R., Eghbali, S., Ramanan, J., Sahota, J., Thakur, S., Wu, S., Smyth, C., Poupart, P., Brubaker, M.: Time2Vec: Learning a vector representation of time. *arXiv* **1907.05321** (2019), [arXiv:1907.05321](https://arxiv.org/abs/1907.05321)
 13. Kim, J., et al.: Large language models are powerful electronic health record encoders. *arXiv* **2502.17403** (2025), [arXiv:2502.17403](https://arxiv.org/abs/2502.17403)
 14. Lee, C., Zame, W., Yoon, J., van der Schaar, M.: DeepHit: A deep learning approach to survival analysis with competing risks. In: *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*. pp. 2314–2321 (2018)
 15. Li, Y., Rao, S., Solares, J.R.A., Hassaine, A., Ramakrishnan, R., Canoy, D., Zhu, Y., Rahimi, K., Salimi-Khorshidi, G.: BEHRT: Transformer for electronic health records. *Scientific Reports* **10**, 7155 (2020). <https://doi.org/10.1038/s41598-020-62922-y>

16. Luo, Y., Li, S., Cao, Z.: GRASP: Group-Shapley feature selection for patients. In: ICASSP 2026 – IEEE International Conference on Acoustics, Speech and Signal Processing. IEEE (2026)
17. Pang, C., Jiang, X., Kalluri, K.S., Spotnitz, M., Chen, R., Perotte, A., Nataraajan, K.: CEHR-BERT: Incorporating temporal information from structured EHR data to improve prediction tasks. In: Proceedings of Machine Learning for Health. Proceedings of Machine Learning Research, vol. 158 (2021), arXiv:2111.08585
18. Rasmy, L., Xiang, Y., Xie, Z., Tao, C., Zhi, D.: Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine* **4**, 86 (2021). <https://doi.org/10.1038/s41746-021-00455-y>
19. Shmatko, A., Jung, A.W., Gaurav, K., Brunak, S., Mortensen, L.H., Birney, E., Fitzgerald, T., Gerstung, M.: Learning the natural history of human disease with generative transformers. *Nature* **647**, 248–256 (2025). <https://doi.org/10.1038/s41586-025-09529-3>
20. Steinberg, E., et al.: Serialized EHR make for good text representations. *arXiv* **2510.13843** (2025), arXiv:2510.13843
21. Team, Q.: Qwen3 technical report. *arXiv* **2505.09388** (2025), arXiv:2505.09388
22. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems. vol. 30 (2017)
23. Wang, C., et al.: Exploring COPD patient clusters and associations with health-related quality of life using a machine learning approach: A nationwide cross-sectional study. *Engineering* **50**, 220–228 (2025)

Appendix: Implementation and Reproducibility Details

Hardware and Runtime. Qwen3-8B embeddings were generated on an NVIDIA RTX 3090 (124,158 patients, ≈ 7 h per embedding run). Bio-ClinicalBERT (110M params) ran on an NVIDIA RTX 4090 at ≈ 30 min per pass. CoxPH models were fitted on CPU with `lifelines` v0.27; evaluation used `scikit-survival` for TD-AUC and IBS. All PCA projections and CoxPH coefficients are fitted on the training split only; validation and test transforms use training-fit parameters.

Code and evaluation scripts: https://github.com/Finley-Maple/Death_Clock.

Feature Dimensions. Table 3 summarises the exact dimensionalities at each stage of the pipeline. The 39-variable count refers to raw clinical variables before any categorical encoding (demographics, 21 biomarkers, 11 binary comorbidity flags, sex, smoking, alcohol status); this is the input width for S1 and the raw-side of concatenation fusion. For summation fusion, categorical variables are independently standardised with the same 39-variable block; the PCA projection is matched to the same 39-dimensional width so that element-wise addition is dimension-consistent.

Table 3. Feature and embedding dimensions by pipeline stage.

Stage	Dimension	Description
Raw feature block	39	Demographics, biomarkers, comorbidities
PCA embedding (concat)	64	Fit on training split only
CoxPH input (S1)	39	Direct raw features
CoxPH input (concat)	103	39 raw + 64 PCA
Summation fusion output	39	PCA matched to raw feature width